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MCP Protocol Development

MCP Protocol Development

Model Context Protocol (MCP) is the core protocol in XiaoZhi AI system for agent interaction with external tools, supporting device control, IoT integration and smart home management.

Protocol Overview

MCP Protocol Features



Protocol Specification

Detailed MCP protocol interaction flow and message structure



Usage Guide

Specific application methods for IoT device control



MQTT+UDP Hybrid

Hybrid communication of MQTT control and UDP audio



Core Functions

Protocol Architecture

- **JSON-RPC 2.0:** Standard communication protocol based on JSON-RPC
- **Tool Registration:** Dynamic registration and discovery of available tools

- **Asynchronous Calls:** Support for asynchronous tool calls and result callbacks
- **Error Handling:** Comprehensive error handling and retry mechanisms

Device Control

- **Smart Home:** Control of lights, air conditioning, appliance devices
- **Sensor Integration:** Temperature, humidity, light, air quality monitoring
- **Scene Automation:** Condition-based automated scene execution
- **Voice Control:** Natural language to device command conversion



Quick Start

Development Environment

1. **ESP32-S3:** Hardware platform requirements
2. **XiaoZhi Firmware:** Firmware version supporting MCP protocol
3. **Network Connection:** Wi-Fi or 4G network support
4. **Development Tools:** ESP-IDF development environment

Basic Usage

1. **Protocol Understanding:** Learn MCP protocol specifications and message formats
2. **Tool Development:** Implement custom device control tools
3. **Integration Testing:** Verify tool registration and call processes

4. **Deployment Application:** Deploy and optimize in real environment

Technical Support:

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